

# Week of August 17, 2026

Boltz-2, Nesso-1, FlashBind, and an FEP+ four-kinase reproduction

Yusheng Cai

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GitHub folder for this week: [weeks/2026-W34](#)

## Executive summary

This report compares three released protein–ligand affinity workflows. Boltz-2 cofolds an atom-resolved complex and predicts affinity; Nesso-1 predicts affinity without coordinate diffusion or an MSA; and, in this benchmark, FlashBind scores a FABind+-predicted ligand pose in an experimental PDB protein structure [1, 2, 3].

I ran all three models on the public 87-compound, four-kinase benchmark and compared predicted scores with experimental potency within each kinase. The weighted Pearson correlations below closely reproduce the paper values. I also evaluated Boltz-2 and the released FABind+ poses against 16 cocrystals; those structure-recovery results are reported separately from affinity ranking.

| Headline results                                       |                                          |                                                                                 |                 |
|--------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------|-----------------|
| Affinity ranking: 87 compounds                         |                                          |                                                                                 |                 |
|                                                        | Weighted Pearson $r$                     | 95% interval                                                                    | Paper reference |
| Nesso-1                                                | <b>0.807</b>                             | 0.731–0.873                                                                     | 0.80            |
| Boltz-2 MSA-1024                                       | <b>0.627</b>                             | 0.484–0.744                                                                     | 0.66            |
| FlashBind, released poses                              | <b>0.533</b>                             | 0.382–0.669                                                                     | 0.53            |
| Structure recovery: median RMSD over 16 cocrystals (Å) |                                          |                                                                                 |                 |
|                                                        | Boltz-2<br>predicted protein<br>+ ligand | FlashBind<br>experimental PDB protein<br>+ FABind+-predicted<br>ligand position |                 |
| Global protein C $\alpha$ , fitted                     | 1.231                                    | 0.272                                                                           |                 |
| Pocket protein C $\alpha$ , fitted                     | 0.400                                    | 0.208                                                                           |                 |
| Ligand heavy atoms after pocket fit                    | <b>0.898</b>                             | <b>0.634</b>                                                                    |                 |

## 1 Benchmark

I used the older OpenFF Git commit `da7c337` because it contains the full 87-example benchmark used in both model papers:  $16 + 16 + 21 + 34 = 87$  neutral protein–ligand examples [4].

Each target defines one separate medicinal-chemistry series: the protein is held fixed while the ligand is changed. The four targets are cyclin-dependent kinase 2 (CDK2), tyrosine kinase 2 (TYK2), c-Jun N-terminal kinase 1 (JNK1), and p38 $\alpha$  mitogen-activated protein kinase. All four are protein kinases, but they are different proteins with different ligand series and different biochemical assays.

| Target / assay series            | Ligands ( $N$ ) | Protein construct input           | Endpoint | Reference PDB |
|----------------------------------|-----------------|-----------------------------------|----------|---------------|
| CDK2 (cyclin-dependent kinase 2) | 16              | CDK2, 303 aa; cyclin A2, 258 aa   | IC50     | 1H1Q          |
| TYK2 (tyrosine kinase 2)         | 16              | TYK2, 302 aa                      | Ki       | 4GIH          |
| JNK1 (c-Jun N-terminal kinase 1) | 21              | JNK1, 370 aa; JIP1 peptide, 11 aa | IC50     | 2GMX          |
| p38 $\alpha$ (MAPK14)            | 34              | p38 kinase, 372 aa                | IC50     | 3FLY          |

Table 1: Frozen benchmark composition.

**Experimental endpoint.** This is the laboratory measurement used as the benchmark’s ground-truth label. IC50 is the concentration that produces 50% inhibition under a particular assay protocol. Ki is an inhibition constant inferred through an enzyme-kinetic model. Smaller IC50 or Ki means stronger apparent inhibition. These endpoints are related but not interchangeable, and neither should automatically be called a binding free energy. Source values in nM or  $\mu$ M were converted to  $\mu$ M and then to  $\log_{10}(\text{value}/\mu\text{M})$  before comparison. CDK2, JNK1, and p38 use IC50 labels; TYK2 uses Ki labels.

## 2 Model description: Boltz-2, Nesso-1, and FlashBind

### 2.1 Boltz-2: cofold and score

At the highest level, Boltz-2 is a biomolecular cofolding model with an additional affinity-prediction pathway. An example input can contain protein sequence(s), ligand chemistry, an MSA, and optional templates or constraints. Its structure trunk integrates these inputs, and a stochastic diffusion decoder generates three-dimensional coordinates for the modeled heavy atoms. The released system returns sampled complex structures and confidence quantities; for protein–ligand inputs it can also return a binder probability and continuous affinity estimates [1]. These learned affinity outputs are assay-calibrated model scores. They are not direct alchemical calculations of an equilibrium  $\Delta G_{\text{bind}}$ .

### 2.2 Nesso-1: predict affinity without coordinate diffusion

Nesso-1 follows a different affinity workflow. It does not run the atom-coordinate diffusion decoder used by Boltz-2, and it supplies protein context through pretrained ESM-2 embeddings instead of an MSA.

The Nesso paper reports more than a tenfold inference speed-up over Boltz-2, and more than twentyfold on some larger targets, while matching or exceeding Boltz-2 on its evaluated affinity benchmarks [2]. Nesso does not return the atom-resolved complex produced by Boltz-2, so the two models serve different structural needs. For this project, I compare their released affinity outputs on identical protein and ligand inputs. Neither affinity head is a molecular simulation or a direct calculation of  $\Delta G_{\text{bind}}$ ; both are learned from experimental potency and affinity labels.

### 2.3 FlashBind: dock then score

At the user-facing level, FlashBind can begin with a protein sequence and a ligand SMILES. Internally, however, the sequence must first be resolved to a three-dimensional receptor. The released preprocessing searches for an experimental or predicted structure and can fall back to a structure predictor when one is unavailable. FABind+ then identifies a candidate pocket and places the ligand in that region [3].

Around this proposed pose, FlashBind constructs a local protein–ligand graph and processes it with an E(3)-equivariant graph neural network. Consequently, rotating or translating the complete coordinate system does not arbitrarily change the prediction. Separate output heads can return a binary activity probability or a continuous assay-derived affinity score. As with the other models, this continuous score is not a thermodynamic calculation of  $\Delta G_{\text{bind}}$ .

This modularity separates pose generation from affinity scoring, which makes large-library screening more practical but also creates a sequential error path:

receptor conformation  $\longrightarrow$  pocket selection  $\longrightarrow$  docked pose  $\longrightarrow$  binding score.

An incorrect receptor or docked pose can therefore limit the scorer even if the final neural network is otherwise well calibrated.

#### Model workflows at a glance

**Boltz-2:** sequence + ligand + MSA  $\rightarrow$  joint trunk + atom diffusion  $\rightarrow$  atom-resolved complex + affinity

**Nesso-1:** sequence + ligand  $\rightarrow$  ESM-2 + pair trunk + predicted pocket  $\rightarrow$  affinity, without a final Cartesian complex

**FlashBind:** 3D protein structure + ligand  $\rightarrow$  FABind+ pocket and pose  $\rightarrow$  local EGNN score

## 3 Benchmark calculation overview

I evaluated the released versions of all three models on the same frozen set of 87 protein–ligand pairs. For each model, I preserved its native affinity output and matched it to the experimental measurement only during the subsequent analysis. The workflows differed mainly in how much structural information they received and whether they generated a new complex.

### 3.1 Nesso-1

I supplied Nesso-1 with the protein sequence and ligand chemistry for each pair. It inferred an interaction region and returned continuous affinity scores without generating a final three-dimensional complex. All 87 pairs completed successfully [2].

### 3.2 Boltz-2

I supplied Boltz-2 with the same protein sequences and ligand chemistry, plus a sequence MSA generated once for each unique protein chain. Boltz-2 produced a sampled atom-resolved complex together with affinity outputs. Because the complete MSAs exceeded the memory of the local 10 GiB GPU, I used the model’s documented 1,024-row MSA subsampling option for every pair. All 87 structure and affinity calculations completed [1].

### 3.3 FlashBind

For FlashBind, I used the authors’ released benchmark inputs: an experimental PDB protein structure and a FABind+-predicted ligand pose for each pair, together with their precomputed molecular representations. I reran the two released affinity scorers and averaged their continuous outputs. I did not rerun the upstream FABind+ pose generation, so this calculation evaluates the released scoring stage rather than the complete docking-to-scoring pipeline. All 87 pairs completed [3].

Exact model revisions, checkpoints, inference settings, commands, hardware records, and output definitions are given in the separate technical methods report.

## 4 Metric policy: affinity and structure

I use two distinct metric families that answer different questions. The **affinity metrics** use all 87 compounds and test agreement between model scores and experimental potency labels. The **structure metrics** use only the 16 pairs with deposited crystal structures and compare Boltz-2 with the released FlashBind/FABind+ poses.

### 4.1 Affinity-ranking metrics on 87 compounds

I calculated the primary statistic in two stages:

$$r_t = \text{Pearson}(y_{t,i}, \hat{y}_{t,i}), \quad \bar{r} = \frac{\sum_t n_t r_t}{\sum_t n_t}.$$

Thus, the comparison asks whether each model ranks compounds correctly *within* each kinase series and then averages the target-level statistics using the compound counts. Spearman  $\rho$  and Kendall  $\tau$  provide complementary rank-based comparisons.

I estimated uncertainty with 2,000 bootstrap replicates (seed 20260817). In each replicate, I resampled compounds with replacement separately inside each target and recomputed the compound-weighted statistic.

### 4.2 Comparison with experimental structures

I identified 16 exact FEP+4 ligand–target identities with deposited crystal structures and used the same reference and metric for FlashBind/FABind+ and Boltz-2.

For each pair, I defined the experimental protein binding pocket as residues having any heavy atom within 5 Å of the ligand. I then fitted the predicted structure to experimental pocket using only matched protein C $\alpha$  atoms. I held that transform fixed while applying it to the predicted ligand and calculated

$$\text{RMSD}_{\text{pose}} = \min_{\pi \in \mathcal{M}} \sqrt{\frac{1}{N} \sum_{i=1}^N \|R\mathbf{x}_i + \mathbf{t} - \mathbf{x}_{\pi(i)}^{\text{ref}}\|^2},$$

where  $R$  and  $\mathbf{t}$  come only from the protein fit and  $\mathcal{M}$  contains complete bond-order- and chirality-compatible atom mappings. The minimum over  $\mathcal{M}$  corrects for chemically equivalent atom permutations.

## 5 Results

| Target            | Pearson $r$  | Spearman $\rho$ | Kendall $\tau$ |
|-------------------|--------------|-----------------|----------------|
| CDK2              | 0.748        | 0.759           | 0.550          |
| TYK2              | 0.917        | 0.921           | 0.767          |
| JNK1              | 0.666        | 0.667           | 0.482          |
| p38               | 0.869        | 0.894           | 0.723          |
| Compound-weighted | <b>0.807</b> | <b>0.819</b>    | <b>0.641</b>   |

Table 2: Nesso-1 released-checkpoint reproduction. The paper reports a compound-weighted Pearson correlation of 0.80 on this benchmark.

| Target            | Pearson $r$  | Spearman $\rho$ | Kendall $\tau$ |
|-------------------|--------------|-----------------|----------------|
| CDK2              | 0.831        | 0.865           | 0.667          |
| TYK2              | 0.807        | 0.779           | 0.617          |
| JNK1              | 0.758        | 0.805           | 0.616          |
| p38               | 0.364        | 0.433           | 0.287          |
| Compound-weighted | <b>0.627</b> | <b>0.666</b>    | <b>0.497</b>   |

Table 3: Boltz-2 MSA-1024 released-checkpoint reproduction. The paper references are weighted Pearson  $r = 0.66$  and Kendall  $\tau = 0.48$ ; the present values are 0.627 and 0.497.

| Target            | Pearson $r$  | Spearman $\rho$ | Kendall $\tau$ |
|-------------------|--------------|-----------------|----------------|
| CDK2              | 0.612        | 0.641           | 0.433          |
| TYK2              | 0.134        | 0.147           | 0.083          |
| JNK1              | 0.546        | 0.578           | 0.425          |
| p38               | 0.677        | 0.591           | 0.413          |
| Compound-weighted | <b>0.533</b> | <b>0.515</b>    | <b>0.359</b>   |

Table 4: FlashBind two-checkpoint ensemble using the authors’ released FABind+ poses. The reported references are weighted Pearson  $r = 0.53$  and Kendall  $\tau = 0.38$ ; the present values are 0.533 and 0.359.

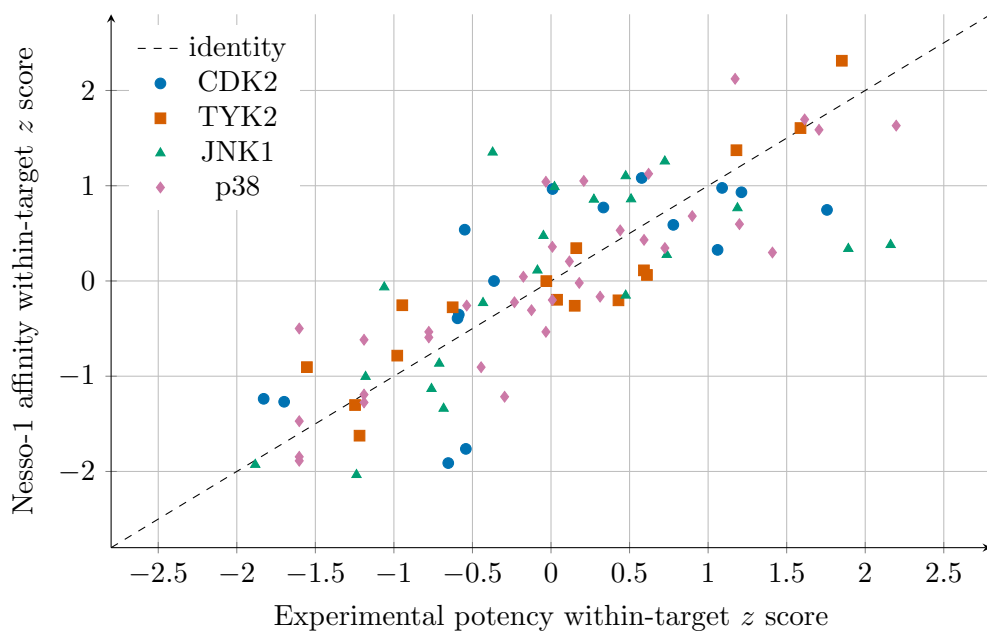


Figure 1: Within-target standardized experimental and predicted potency. Lower values correspond to stronger binding on both axes. Standardization removes target-specific offsets and scale differences for visualization only; metrics were computed from the unstandardized values.

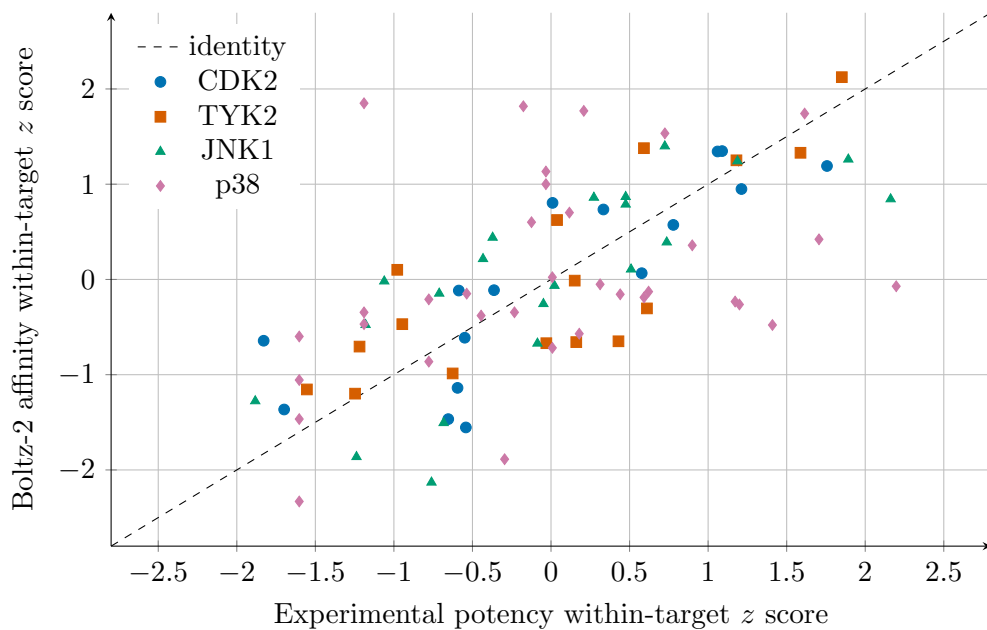


Figure 2: Boltz-2 MSA-1024 standardized scores. The dispersed p38 series is the main target-level departure from Nesso-1 and from the other Boltz targets.

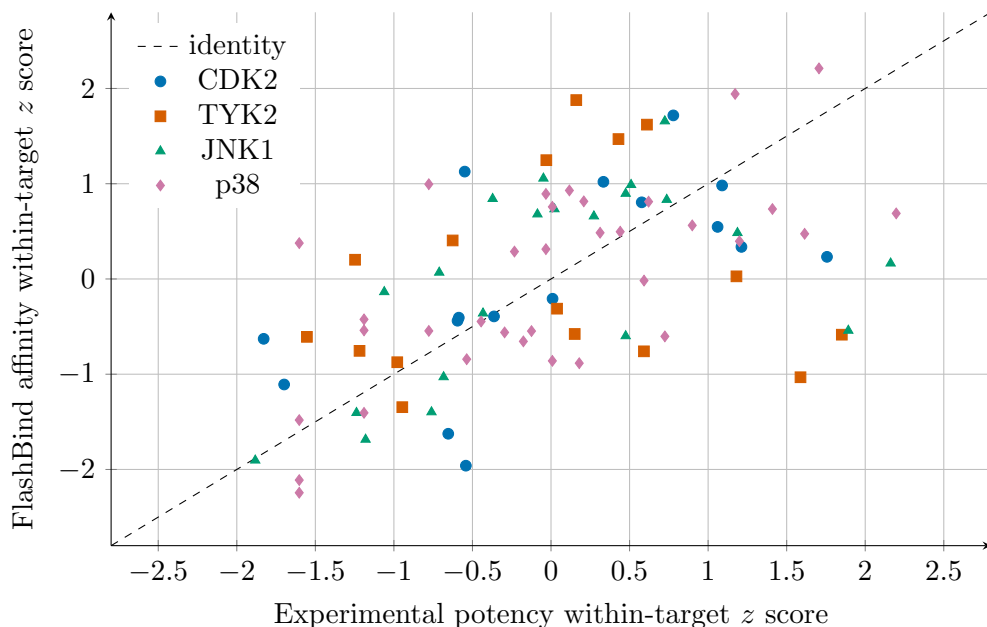


Figure 3: FlashBind standardized scores from the released FABind+ poses. TYK2 is the clear weak target, whereas the other three series retain positive within-target ranking relationships.

I obtained an aggregate Pearson result of 0.807 for Nesso (95% interval 0.731–0.873), which agrees closely with the reported 0.80. For Boltz, I obtained 0.627 (interval 0.484–0.744), which is 0.033 below the paper reference of 0.66; its Kendall result of 0.497 is 0.017 above the paper reference of 0.48. Given reconstructed preprocessing, one random seed, and an explicit MSA-depth change, I regard this as a close released-model reproduction rather than numerical identity.

For FlashBind I obtained weighted Pearson  $r = 0.533$  (95% interval 0.382–0.669), essentially identical to the reported 0.53. The weighted Kendall value was 0.359, 0.021 below the reported 0.38. The close Pearson match is reassuring for software reproduction, but it is not independent validation: the authors supplied the FABind+ poses and representations used by their own scoring benchmark.

| Statistic       | Nesso-1      | Boltz-2      |              |
|-----------------|--------------|--------------|--------------|
|                 |              | MSA-1024     | FlashBind    |
| Pearson $r$     | <b>0.807</b> | <b>0.627</b> | <b>0.533</b> |
| Spearman $\rho$ | 0.819        | 0.666        | 0.515        |
| Kendall $\tau$  | 0.641        | 0.497        | 0.359        |

Table 5: Full matched comparison across the 87-compound benchmark. Each entry is the compound-count-weighted mean of the corresponding within-target statistic.

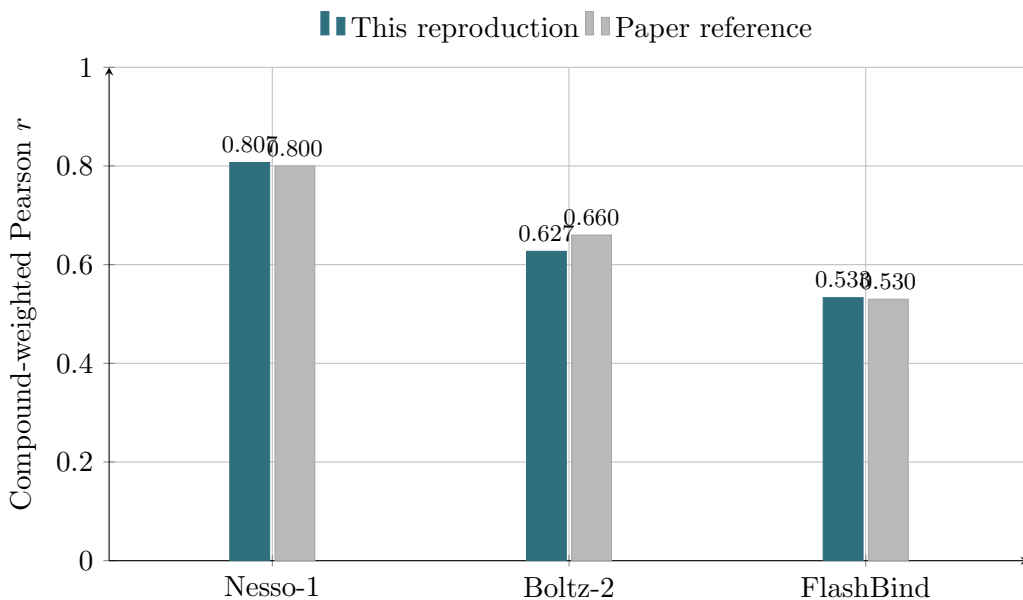


Figure 4: Weighted within-target Pearson correlations from this reproduction and the corresponding paper reference values. The reproduced values differ from the references by +0.007 for Nesso-1,  $-0.033$  for Boltz-2, and +0.003 for FlashBind. Bars show point estimates; the bootstrap intervals are reported in the text and executive-summary table.

I found that the paired interval excludes zero for all three correlation statistics, supporting a difference in within-target ranking performance. Target-level heterogeneity is scientifically important: Boltz is better on CDK2 and JNK1, Nesso is better on TYK2, and p38 supplies the largest Nesso advantage ( $r = 0.869$  versus  $0.364$ ). My experiment does not determine whether that p38 difference arises from model architecture, MSA subsampling, checkpoint stochasticity, or reconstructed input/preprocessing details. FlashBind shows a different pattern: p38 is its strongest target ( $r = 0.677$ ), while TYK2 is much weaker ( $r = 0.134$ ). Because this run uses released docked poses, that TYK2 result cannot be diagnosed as a fresh docking failure without separately examining the archived poses.

## 5.1 Protein and ligand RMSD on the 16 structure-matched pairs

| RMSD component ( $\text{\AA}$ )     | Boltz-2                    |       | FlashBind                                                       |       |
|-------------------------------------|----------------------------|-------|-----------------------------------------------------------------|-------|
|                                     | predicted protein + ligand |       | experimental PDB protein +<br>FABind+-predicted ligand position |       |
|                                     | Median                     | Mean  | Median                                                          | Mean  |
| Global protein $C_\alpha$ , fitted  | 1.231                      | 1.282 | 0.272                                                           | 0.279 |
| Pocket protein $C_\alpha$ , fitted  | 0.400                      | 0.998 | 0.208                                                           | 0.220 |
| Ligand heavy atoms after pocket fit | 0.898                      | 0.977 | 0.634                                                           | 0.651 |

Table 6: Separate protein and ligand errors across the same 16 cocrystal references. No joint protein–ligand RMSD is used. The global and pocket protein rows are residual  $C_\alpha$  RMSDs after fitting the indicated protein atoms. The ligand row is symmetry-corrected heavy-atom RMSD after applying the fixed pocket-protein transform; the ligand is not independently fitted.

For Boltz-2, the protein rows evaluate its predicted protein structure. For FlashBind, the protein coordinates come from an experimental PDB, while the ligand position and orientation are predicted by FABind+. Its protein rows therefore measure differences among experimental protein structures, not FlashBind protein



prediction accuracy. The separate rows avoid allowing the much larger protein atom set to dominate a combined RMSD and avoid using the experimental ligand coordinates to improve the fit.

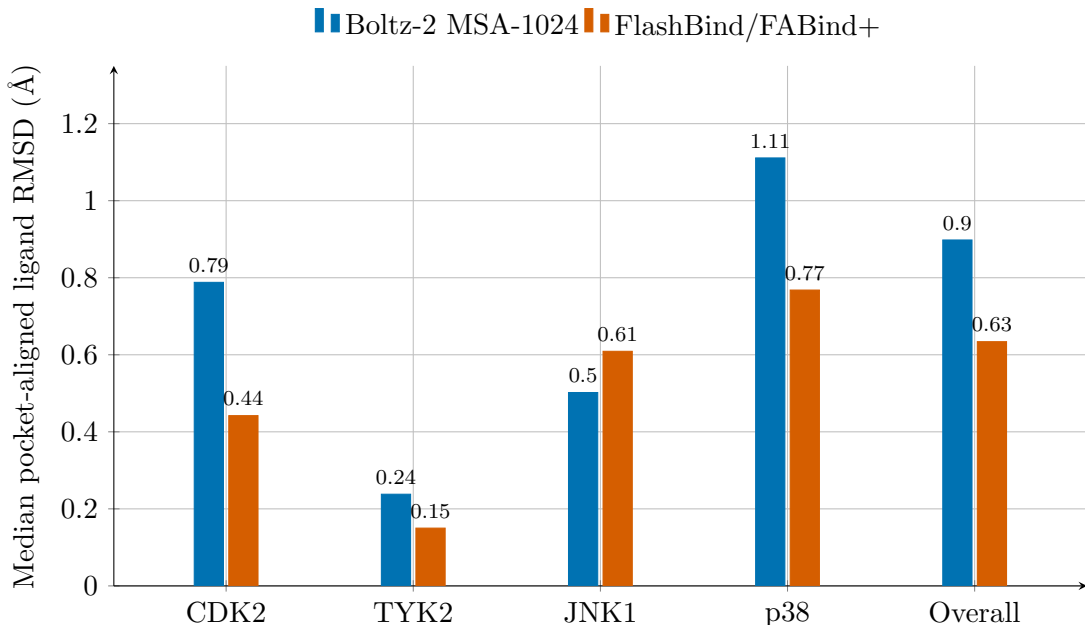


Figure 5: Median symmetry-corrected ligand RMSD by target and over all 16 structure-matched pairs. The target counts are CDK2  $n = 6$ , TYK2  $n = 2$ , JNK1  $n = 1$ , and p38  $n = 7$ ; the JNK1 bar is therefore a single pair rather than a stable target-level estimate. FlashBind has the lower overall median, but the per-pair comparison remains mixed (9 lower for FlashBind and 7 for Boltz-2).

The released FlashBind/FABind+ poses were below 2 Å for all 16 pairs, whereas the Boltz-2 poses were below 2 Å for 14. FlashBind had the lower RMSD on 9 pairs and Boltz-2 on 7. The median paired FlashBind-minus-Boltz difference was  $-0.109$  Å; its mean was  $-0.326$  Å, influenced by the two Boltz poses above 2 Å. This pattern does not support a claim of uniform FlashBind superiority.

The protein decomposition changes the interpretation. Boltz-2’s median global protein  $C\alpha$  RMSD was 1.231 Å, while its median locally fitted pocket  $C\alpha$  RMSD was 0.400 Å. The pocket result was strongly target-dependent: the Boltz medians were 0.348 Å for CDK2, 0.243 Å for TYK2, 0.460 Å for JNK1, and 2.305 Å for p38. Thus the p38 ligand-pose result coincides with a substantial local pocket-structure error and should not be interpreted as ligand placement alone. In contrast, the FlashBind protein pocket median was 0.208 Å overall because it started from experimental PDB protein structures.

I also audited the coordinate identities before scoring. Every released SDF matched its reference ligand in heavy-atom count and admitted a complete chemical graph mapping. The released p38 receptor has a formatting defect in which the chain identifier disappears after residue 168; I joined the two continuously numbered fragments in memory without changing coordinates and recorded the repair. The archive documents these ligands as FABind+ docking poses, but it does not contain the original docking commands, seeds, or per-pose logs. Thus I have evaluated the supplied poses, not independently reproduced their generation.

## 6 Conclusion

The three models address protein–ligand affinity in distinct ways. Boltz-2 uses protein sequence, ligand chemistry, and an MSA to generate an atom-resolved complex and predict affinity. Nesso-1 predicts affinity from protein sequence and ligand chemistry without generating a final Cartesian complex. FlashBind instead scores a proposed three-dimensional complex; in this benchmark, that complex consisted of an experimental PDB protein structure and a FABind+-predicted ligand pose.

I reproduced the headline affinity-ranking result reported for each model on the same 87-compound benchmark. The compound-weighted Pearson correlations were 0.807 for Nesso-1, 0.627 for Boltz-2, and 0.533 for FlashBind, compared with paper values of 0.80, 0.66, and 0.53, respectively. The absolute differences were therefore 0.007, 0.033, and 0.003. This close agreement shows that the released models and the analysis used here reproduce the central within-target ranking results from the three papers.

This agreement is a reproducibility result rather than an independent test of generalization: all calculations use the same public benchmark, and the three workflows do not receive equivalent structural information. It nevertheless provides a validated starting point for studying how their architectures and inputs produce affinity predictions.

## 7 Next week’s agenda

Next week, I will move from reproducing benchmark results to understanding how the three models produce those results. I will organize the architectures in a common sequence—input representation, interaction module, structural or pocket information, and affinity readout—so that their design choices can be compared directly.

Using one protein–ligand pair as a worked example, I will trace the information passed through each released implementation and address three questions:

1. How are the protein, ligand, MSA, and three-dimensional coordinates represented?
2. How does each model combine protein and ligand information before the affinity head?
3. Which controlled input changes can reveal what the prediction depends on, such as MSA depth for Boltz-2, protein-sequence changes for Nesso-1, and ligand-pose perturbations for FlashBind?

The intended outputs are a common architecture diagram and a small notebook that follows one example through the major stages of each model.

## Technical methods

Exact source revisions, checkpoint checksums, resolved inference settings, commands, hardware records, output fields, and validation procedures are documented separately in the [Week 34 technical methods report](#). This presentation report retains only the procedural detail needed to interpret the scientific comparisons.

## References

- [1] Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi, Julien Roy, Hannes Stark, David Kwabi-Addo, Dominique Beaini, Tommi

- Jaakkola, and Regina Barzilay. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707. URL <https://doi.org/10.1101/2025.06.14.659707>. Preprint, version 1, posted 18 June 2025.
- [2] Nikhil Shenoy, David Errington, Emmanuel Bengio, Kacper Kapuśniak, Kerstin Klaeser, Yui Tik Pang, Vladimir Radenkovic, Prudencio Tossou, Therence Bois, Andrew Wedlake, and Francesco Di Giovanni. Nesso-1: Accelerating open-source binding affinity predictions. *bioRxiv*, 2026. doi: 10.64898/2026.08.01.742196. URL <https://doi.org/10.64898/2026.08.01.742196>. Preprint, version 1, posted 3 August 2026.
- [3] Songlin Jiang, Yifan Chen, Aarti Krishnan, Yu Zhang, and Wengong Jin. Flashbind: Towards accurate and efficient structure-based virtual screening. *bioRxiv*, 2026. doi: 10.64898/2025.12.22.695983. URL <https://doi.org/10.64898/2025.12.22.695983>. Preprint, version 2, posted 8 April 2026; version 1 was titled FlashAffinity.
- [4] David F. Hahn et al. Best practices for constructing, preparing, and evaluating protein-ligand binding affinity benchmarks. *Living Journal of Computational Molecular Science*, 4(1):1497, 2022. doi: 10.33011/livecoms.4.1.1497. URL <https://doi.org/10.33011/livecoms.4.1.1497>.